

SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

Online Examination Management System (OEMS)

Submitted By:

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BS Computer Science (8th Semester)

Project Title:

Online Examination Management System (OEMS)

1. Introduction

The Online Examination Management System (OEMS) is a web-based application developed to automate and manage examination activities in educational institutions. The system enables administrators, teachers, and students to perform examination-related tasks through a centralized platform.

Frontend Technologies:

- HTML
- CSS
- JavaScript

Backend Technologies:

- PHP

Database:

- MySQL
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2. System Requirements

Hardware Requirements

Server Side

- Processor: Intel Core i3 or above

- RAM: Minimum 4 GB
- Hard Disk: 100 GB Available Space
- Internet Connection

Client Side

- Computer/Laptop/Smartphone
 - Web Browser (Chrome, Firefox, Edge)
 - Internet Connection
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Software Requirements

- Operating System: Windows 10/11 or Linux
 - Web Server: XAMPP/WAMP
 - PHP 8.0 or Above
 - MySQL Database
 - HTML, CSS, JavaScript
 - Visual Studio Code
 - Web Browser
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3. User Requirements

The system shall provide the following user roles:

Administrator

- Manage teachers and students
- Manage courses and subjects
- Schedule examinations
- Generate reports
- Monitor system activities

Teacher

- Create question bank
- Create and manage exams
- Evaluate subjective answers
- View student performance
- Publish results

Student

- Login securely
 - View available exams
 - Attempt online examinations
 - View results and grades
 - Update profile information
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4. Functional Requirements

FR-01 User Authentication

- Users must login using username and password.
- Role-based access control shall be implemented.

FR-02 User Management

- Admin can add, edit, update and delete users.
- Bulk student upload through CSV file.

FR-03 Question Bank Management

- Add new questions.
- Edit existing questions.
- Delete questions.
- Categorize questions by subject and difficulty level.

FR-04 Exam Creation

- Teachers can create exams.

- Set exam duration.
- Assign total marks.
- Select questions manually or randomly.

FR-05 Exam Scheduling

- Set exam date and time.
- Control exam availability.

FR-06 Online Examination

- Students can attempt exams online.
- Countdown timer will be displayed.
- Automatic submission when time expires.

FR-07 Auto Save

- Student answers will be saved automatically.
- Prevent data loss due to internet interruption.

FR-08 Result Generation

- Automatic checking of MCQs.
- Manual grading for descriptive questions.
- Generate result reports.

FR-09 Report Management

- Generate PDF reports.
- Export results to Excel.
- View student performance statistics.

FR-10 Notification System

- Notify students about upcoming exams.
- Notify students when results are published.

5. Non-Functional Requirements

Security

- Password encryption.
- Secure login system.
- Role-based permissions.

Performance

- Support multiple users simultaneously.
- Fast page loading.

Reliability

- Automatic data backup.
- Consistent database transactions.

Availability

- System available 24/7 during examination periods.

Usability

- User-friendly interface.
- Easy navigation for all users.

Compatibility

- Compatible with Chrome, Firefox and Edge browsers.
- Responsive design for mobile and desktop devices.

6. Database Requirements

The system shall maintain the following tables:

- Users
- Students
- Teachers
- Subjects
- Courses

- Question Bank
 - Exams
 - Exam Questions
 - Student Answers
 - Results
 - Notifications
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7. Expected Outputs

- Online examination portal
 - Automated grading system
 - Student result management
 - Question bank management
 - Examination reports
 - Secure user authentication system
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8. Conclusion

The Online Examination Management System will provide a secure, efficient and automated environment for conducting examinations. The system will reduce manual work, improve accuracy, save time and provide quick access to examination results for educational institutions.